Three Further Lean Proof Simplifications

For ProveIt's DiophantineConsistency development

A sharp rank bound, a two-equation zero test, and an elementary unit proof

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Abstract

This review proposes three residual simplifications after treating all ideas in 43 earlier proposal packets as available. The principal arithmetic result is the sharp upper bound $R \leq q^4 - q^3 - q - 1$ for the reduced native-rank constraints, proved by an explicit sum of nonnegative terms. It replaces a sequence of coarser estimates in `source_window_bound`. A second result proves the zero-test selector from annihilation and balance alone, removing the Boolean idempotence argument from two existing soundness proofs while keeping the literal certificate unchanged. A smaller third change proves that the actual controller base is a unit from oddness of the modulus, without an element-order calculation. The accompanying proof-only patches remove 30 physical lines from four current modules, with no new production helpers or imports. The mathematical kernels and source-shaped verification files were checked with Lean 4.32.0 and the repository's pinned mathlib revision. A complete rebuild of the production import graph was not performed; that remaining integration check is explicitly separated from the checks completed here.

1 Scope, baseline, and what is new

The reviewed project is `Computability/DiophantineConsistency` in the ProveIt repository. The initial source revision was `3299f4aa4c9a`. During the review, the exclusion corpus grew from 41 to 43 packets. The final exclusion revision is `60fcb2a69b7f`; the intervening three commits add the two packets `2026-10-26zp` and `2026-10-26zq` and update the two indexes. They do not change maintained Lean source. The full revision identifiers and source hashes are recorded in the evidence directory.

A final read-only freshness check found main at `09eaa57ae5b8` (26 October, 19:20:44 UTC). Its nine additional commits add further construction results and adopt previously excluded small cleanups. The four target files in this packet are unchanged; the proposal collection still has 43 packets. The only change inside the requested exclusion directories is an adoption-note update for existing list-lookup and predecessor-power ideas. The final comparison is preserved in `evidence/final_freshness.json`.

The user's exclusion rule is interpreted semantically. An earlier idea counts as prior work even if it was uncompiled, deferred, rejected, described only as a future direction, or never adopted. This rules out repackaging an old helper, compiling an earlier sketch, or moving a theorem to a more convenient file as a new proposal. It also rules out counting a shorter proof already proposed for another client in an explicitly covered family.

The audit combined the packet index, adoption notes, prior exclusion maps, manuscript headings, recent packet READMEs and reports, and candidate-specific searches across prose, Lean files, and patches. This is a documented comparison, not a claim that a textual search mechanically proves mathematical novelty. The candidate decisions identify the closest earlier mechanisms and why the retained proposals differ. For example, a tempting simplification of `CurrentCellLabels2011.sum_le_one_cases` was rejected: the exact `Finset.sum_le_one_iff` support characterization occurs in several earlier packets, even though it was absent from one convenient summary ledger.

Proposal	Existing targets	Distinct mechanism
Sharp upper bound	`NativePellBootstrap.source_window_bound`	Replace independent estimates by a sharp nonnegative-deficit identity.
Two-equation zero test	`QuadraticBoolean.zero_certificate_sound`; `ObjectZeroTest.certificate_sound`	Derive the selector without the idempotence premise.
Controller base unit	`CurrentControllerSubsets2011.block_base_isUnit`	Use oddness and a power of two directly; avoid the order theorem in this proof term.

The first two are the main recommendations. The third is deliberately a small, local dependency improvement. No new encoding, variable count, equation, certificate row, or consistency result is claimed.

2 A sharp upper bound for the native rank

2.1 The existing proof obligation

The current `source_window_bound` proves

$$\mathrm{nativeRank}(x) + 1 < x_{17}$$

from positive source coordinates, $\beta > 0$, $K \geq 0$, and the two upper mask bounds. Its existing proof estimates a quadratic expression G, estimates the two mask contributions separately, combines those estimates, proves a further comparison of powers, and finally compares the result with the source scale. Those steps are visible in the pinned source declaration.

Write the relevant source coordinates as

$$J = x_0, \qquad F = x_1, \qquad \alpha = x_2, \qquad \beta = \mathrm{ports}_0, \qquad q = 1 + J\beta,$$

and let $M_C = \mathrm{ports}_2$, $M_F = \mathrm{ports}_3$, $R = \mathrm{nativeRank}(x)$, and $E = (x_{17} : \mathbb{Z})$. The already established source identities give

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \tag{1}$$

$$E = q^4 ws, \qquad w = x_{10} \geq 1, \quad s = x_9 \geq 1. \tag{2}$$

The existing positive-transport theorem gives $q - F - \alpha > 0$. Thus, over integers, the arithmetic assumptions relevant to the upper bound are

$$F, \alpha, J \geq 1, \quad C := q - F - \alpha \geq 1, \quad M_C \leq \beta - 1, \quad M_F \leq 2\beta - 1. \tag{3}$$

In particular $q = F + \alpha + C \geq 3$. No lower bound on either mask port is needed for this upper estimate. This agrees with the actual theorem: it does not assume the full `PortBox` record.

2.2 The exact deficit

Proposition 1 (Sharp reduced upper bound)**.** *If (1), (3), and $q = 1 + J\beta$ hold in $\mathbb{Z}$, then*

$$\boxed{R \leq q^4 - q^3 - q - 1.}$$

Proof. The following polynomial identity is the entire substantive estimate:

$$\begin{aligned} q^4 - q^3 - q - 1 - R = {} & q(q^2-1)(F-1) \\ & + (q^2-1)(q-F-\alpha-1) \\ & + (q+1)(J-1) \\ & + J\big((\beta - M_C - 1) + q(2\beta - M_F - 1)\big). \end{aligned} \tag{4}$$

After substituting $q = 1 + J\beta$, Lean's `ring` tactic checks the identity. Every summand on the right is nonnegative: $q \geq 3$, $q^2 - 1 \geq 0$, and each remaining factor is one of the nonnegative slacks in (3). Therefore the deficit is nonnegative. □

The important change in proof organization is that the mask terms are not estimated independently and then recombined with a loose bound on G. Equation (4) keeps all the integrality margins until the end. Each hypothesis has a visible role in a particular nonnegative factor. This also makes future changes to a mask threshold easier to assess: they change specific terms of the certificate.

2.3 Why the bound is sharp, and what that does not imply

For any integer $q \geq 3$, set

$$J = 1, \quad \beta = q - 1, \quad F = 1, \quad \alpha = q - 2, \quad M_C = q - 2, \quad M_F = 2q - 3.$$

These values satisfy the reduced constraints, and every term in (4) is zero. Hence equality holds. Both the equality and the admissibility of this family are proved in the delivered Lean file.

This establishes sharpness for the reduced arithmetic box only. It does not assert that these values extend to a positive solution of every source row, the Pell equations, or the final Diophantine equation. The source has additional constraints that were deliberately not used here. A smaller bound could still follow from those extra constraints.

The strict integer hypotheses matter. The conversion of, for example, $M_C < \beta$ into $\beta - M_C - 1 \geq 0$ uses integer discreteness. The proposition is not stated for arbitrary real variables under the same strict inequalities. It also retains the positive F, α, J and transport-gap hypotheses; these are not convenient assumptions silently introduced by the proof.

2.4 Recovering the existing window theorem

Equation (2) gives $q^4 \leq E$. The new bound then gives

$$R + 1 \leq q^4 - q^3 - q < q^4 \leq E,$$

because $q \geq 3$. Thus the existing strict window statement follows with room to spare. The original theorem's type and downstream uses stay identical.

The supplied production patch inserts the deficit argument directly into `source_window_bound`. It adds no helper declaration and no import. The separate general theorem is included as a reusable mathematical artifact, not as an uncharged production dependency. The patch explicitly unfolds `nativeTarget` when checking the identity; it does not rely on accidental unfolding by `ring`.

The current declaration changes from 50 to 36 physical lines. More significantly, the separate G, mask, recombination, and $q^2 < q^3$ estimates are replaced by one equality and nonnegativity of its

right-hand side. There is no claimed elaboration-time improvement: the new `ring` obligation is larger, and timing would require a controlled full module comparison.

2.5 Relation to earlier proposals

Packet `2026-10-26u` already proposes a *lower* native-rank envelope, including $(2q-1)(q^2-1)$. Packet `2026-10-26zb` compares ranks using the scalar $C+qF$ at fixed surrounding parameters. Both are treated as available. Neither supplies the upper deficit (4), its sharp reduced-box bound, or this replacement of `source_window_bound`. The new statement therefore complements those results rather than reimplementing them.

3 Prove the zero test without the Boolean equation

3.1 The certificate already contains enough information

The current internal certificate has three fields:

$$b^2 = b, \qquad bn = 0, \qquad n + b(s+1) = s+1. \tag{5}$$

Its soundness proof first applies the internal Boolean-root theorem to the first field, then handles $b = 0$ and $b = 1$. The external natural-coordinate certificate uses the integer rows

$$b(b-1) = 0, \qquad bn = 0, \qquad n - (1-b)(s+1) = 0$$

and follows the same strategy. The relevant pinned declarations are `ObjectZeroTest.certificate_sound` and `QuadraticBoolean.zero_certificate_sound`.

The first field is not needed to determine the selector.

Proposition 2 (Annihilation and balance determine the selector)**.** *Let A be a commutative semiring with cancellation by nonzero factors, let $u \neq 0$, and suppose*

$$bn = 0, \qquad n + bu = u.$$

Then $b = 1$ if $n = 0$, and $b = 0$ if $n \neq 0$.

Proof. If $n = 0$, the balance equation becomes $bu = u = 1u$. Cancel the nonzero right factor u to obtain $b = 1$. If $n \neq 0$, the annihilation equation and absence of zero divisors give $b = 0$. □

This proof does not first establish that b is Boolean. Booleanity is a consequence of the selector equation. For the internal certificate take $u = s+1$ and use the existing theorem that this internal successor is nonzero. For the external certificate, rearrange the third integer row into the balance equation and use that a natural slack plus one is positive.

3.2 The actual proof edits

The production changes retain the definitions of both certificates and all three literal polynomial rows. They change only the two soundness proof bodies. In each proof, the first field is explicitly cleared, and the case split is on $n = 0$. The remaining equations then simplify directly.

Explicit clearing is a useful implementation detail here. An underscore in a destructuring pattern may leave an automatically named hypothesis available to automation. Since this proposal is specifically about not using the first field, the delivered proof uses the clearing pattern and is checked after that change. The independent general theorem has no idempotence hypothesis at all, which makes the logical reduction unambiguous.

The external body changes from 10 to 3 lines, and the internal body from 9 to 3 lines. Together these direct edits remove 13 physical lines without adding a shared helper or an import. A general shared theorem is available in the standalone artifact if future clients justify it, but its code is not hidden from the production accounting.

3.3 Why this works for nonstandard internal naturals

The argument is algebraic. It does not replace the internal carrier with $\mathbb{N}$, enumerate its values, or perform external induction on its elements. The existing internal semiring, linear order, canonical additive order, and strict ordered-ring structure provide cancellation by a nonzero factor. The canonical additive order supplies the `ExistsAddOfLE` premise needed by mathlib's ordered-domain instance. The existing `ObjectNaturalDiscrete.add_one_ne_zero` supplies the nonzero successor fact for arbitrary internal slack values. The delivered abstract-carrier fixture checks the argument under these algebraic contracts.

There are genuine hypotheses here. If $u = 0$ and $n = 0$, both equations hold for every b, so the nonzero condition cannot be dropped. Likewise, arbitrary semirings with zero divisors do not support the conclusion. For example, in $\mathbb{N} \times \mathbb{N}$, choose $n = (1, 0)$, $b = (0, 1)$, and $u = (1, 1)$. Annihilation and balance hold, but b is neither the zero nor the unit of the product semiring. The report therefore does not advertise the result as a theorem of all commutative semirings.

3.4 Relation to earlier proposals and to equation size

The nearest earlier work includes packet `2026-10-26m`'s use of the zero-test correctness theorem in conditional clients, packet `2026-10-26x`'s simplification of idempotent cancellation, and packet `2026-10-26zg`'s split-zero interpretation of signed expressions. These are useful but different: they still consume Booleanity, simplify how it is proved, or normalize the interpretation of rows. The present argument shows that the other two constraints already force the selector.

Removing the literal Boolean row could be a separate encoding experiment. It is not part of this patch: doing so would alter the generated system and require new row, degree, operation-count, and representation checks. The delivered proposal is only a proof simplification and a precise redundancy theorem. All public statements, witness counts, and executable expressions remain unchanged in the proposed files.

4 Use oddness to prove the controller base is a unit

4.1 The existing endpoint needs less than an order theorem

The current `block_base_isUnit` obtains positive order of B^{a-1} from `block_order`, infers that this power is a unit, and then deduces that B is a unit. For this particular conclusion, the established

numeric shape of the base and modulus is enough:

$$B = 2^{94517}, \qquad d = 47 \cdot 2011, \qquad a = 120199, \qquad m = (da)a^k.$$

The existing controller-data module already provides `B_eq`, `d_odd`, and `auxiliary_odd`.

Since m is odd, $\gcd(2, m) = 1$. Hence the residue class of 2 is a unit in $\mathbb{Z}/m\mathbb{Z}$, and so is its 94517th power. The proof works also at $k = 0$; there is no positive-exponent side case for k.

In Lean, the useful chain is

```
Nat.coprime_two_left
ZMod.isUnit_iff_coprime
IsUnit.pow
```

with `Nat.cast_pow` moving the exponentiation into `ZMod`. This keeps 2^{94517} symbolic. An earlier test form applied coprimality to the large natural power and triggered an elaborator warning about reducing the large exponent; the supplied final proof avoids that form.

The declaration changes from 8 to 5 physical lines. The claim is strictly local: the proof term no longer uses `block_order`. The surrounding module still needs its power-coordinate imports for other results. There is no claim that the whole module's import graph, total build time, or all controller order calculations disappear.

4.2 Novelty boundary

Previous packets contain general coprime-thickening, fixed-reference Gamma certificates, exact-order alternatives, and weaker controller annihilation facts. Those mechanisms are excluded. This proposal is only the remaining endpoint replacement for the actual base B. In particular, `2026-10-26zq/evidence/late_head_scope_addendum.md` explicitly leaves the base-unit proof in use after its own subset construction. The present small change can therefore coexist with that proposal.

5 Validation, accounting, and integration

5.1 Completed checks

The checked runtime reports Lean 4.32.0, commit `8c9756b28d64dab099da31a4c09229a9e6a2ef35`. Its mathlib source is `81a5d257c8e410db227a6665ed08f64fea08e997`, matching the repository's `lake-manifest.json`. An existing installation was reused read-only; project object files from other checkouts were not used for these checks.

The delivered Lean files check the general arithmetic identities, their inequalities, the sharpness family, the selector theorem, the literal external certificate, an abstract admissible internal certificate, and the actual numerical controller modulus. Source-shaped fixtures additionally check the proposed rank body with the copied rank definitions and the existing source identities passed as explicit hypotheses, and check the controller body under the existing base-identity and oddness contracts.

These hypotheses are not `axiom` declarations or admitted proofs. They are theorem parameters representing the already proved production interfaces. This checks how the proposed body consumes

those interfaces. It does not recompile the proofs that establish the interfaces in the complete project. The source manifest and review identify the exact production suppliers.

All checked files avoid `sorry`, `admit`, and added axioms. Their `#print axioms` outputs contain only the ordinary logical foundations `propext`, `Classical.choice`, and `Quot.sound`, as applicable. The logs are included. This is a check of these new lemmas and fixtures; it should not be read as an axiom audit of every production theorem in ProveIt.

5.2 What the line counts mean

Changed module	Before	After	Removed
`NativePellBootstrap`	671	657	14
`QuadraticBoolean`	136	129	7
`ObjectZeroTest`	91	85	6
`CurrentControllerSubsets2011`	102	99	3
Total reduction			30

These are physical-line differences for the supplied complete replacement files against the pinned current source. New production helpers and imports are zero. The extra standalone proofs, fixtures, report, and audit scripts are validation and explanatory artifacts; they are not proposed production dependencies. Declaration types and all non-proof definitions are preserved. One explanatory controller comment is updated to describe its new proof.

The numbers are not an estimate against an imaginary repository in which every alternative historical patch has simultaneously been merged. Some historical alternatives are mutually exclusive. The user's all-applied rule is enforced for novelty; quantitative differences are measured against an explicit real revision. This distinction avoids charging the same earlier improvement twice.

5.3 Remaining production check

A complete build of the four production modules and their transitive project dependencies was not performed here. The initial environment had very little free storage, and no fresh full project build was claimed in its place. The checked kernels and source-shaped fixtures establish a narrower, useful verification boundary. The report consistently labels the patches as integration candidates.

The archive provides scripts for source verification and standalone checks with `lake env lean`. The README also gives commands to apply the patches and build the four changed targets and their relevant clients in a ProveIt checkout. The patches do not require a new mathematical assumption. If the current repository has moved, compare the target bodies and signatures before applying them; do not force a patch through a conflicting change.

Recommended order is the zero-test change first, because it is small and its logical premise reduction is especially easy to review; the rank change next, because it supplies the most substantive new identity; and the unit proof last as a small cleanup. These changes are independent and can be accepted separately. No GitHub commit, pull request, or repository write was made as part of this delivery.

6 Further research questions

These are follow-up questions suggested by the three results, not additional claims of novelty.

Can the exact source improve the sharp reduced bound? The reduced-box extremizer need not satisfy the Pell and transport equations. Does an additional source row exclude it, and if so, can that exclusion be expressed as another positive term in (4)? A useful result would be a new source-specific margin with a short certificate, not a larger collection of numerical estimates.

Should the upper envelope become public? The current patch keeps the argument local to avoid helper overhead. If a second independent client needs a bound on the native rank, the separately checked `upper_envelope` is a candidate public interface. Its precise strength and mask-sign assumptions are already documented. The decision should be based on real consumers and total code cost.

Where else is an algebraic certificate premise redundant? The zero-test example suggests inspecting certificates for constraints implied by their remaining equations and nonzero guards. This is a research direction, not a claim that arbitrary rows can be removed. Each candidate needs a proof over the actual carrier, including nonstandard internal values, and its literal encoding should remain fixed during proof-only refactoring.

Source and artifact guide

All repository references are pinned to the final exclusion revision unless otherwise stated. The evidence directory includes the candidate-specific novelty comparison, independent mathematical review, source hashes, detailed line metrics, runtime validation results, and logs. The complete changed files are supplied alongside patches, so a reviewer can inspect either form.

1. Prior proposal corpus and index, including all 43 packets.
2. Incoming index; at the reviewed revision it contains no waiting archive.
3. Native rank, target, scale, and window theorem.
4. External quadratic Boolean certificate and internal zero-test certificate.
5. Controller constants and parity facts and the base-unit consumer.
6. Pinned mathlib source; relevant library contracts are checked by the delivered Lean files.